

EXERCISE-1

Q1. Setup Postgres Install any latest stable version. Run using the terminal and create a database and table in that.

-To install postgres go to : <https://www.postgresql.org/download/> and follow the steps mentioned there.

Create a database:

```
postgres=# CREATE DATABASE exercise;
CREATE DATABASE
postgres=# \l
postgres=# \l
```

List of databases									
Name	Owner	Encoding	Locale Provider	Collate	Ctype	ICU Locale	ICU Rules	Access privileges	
exercise	postgres	UTF8	libc	en_US.UTF-8	en_US.UTF-8				
postgres	postgres	UTF8	libc	en_US.UTF-8	en_US.UTF-8				
practice_db	postgres	UTF8	libc	en_US.UTF-8	en_US.UTF-8				
template0	postgres	UTF8	libc	en_US.UTF-8	en_US.UTF-8			=c/postgres	+
								postgres=CTc/postgres	
template1	postgres	UTF8	libc	en_US.UTF-8	en_US.UTF-8			=c/postgres	+
								postgres=CTc/postgres	

(5 rows)

A table in that database:

```
exercise=# CREATE TABLE students(
exercise=# student_id SERIAL PRIMARY KEY,
exercise=# name TEXT NOT NULL,
exercise=# email TEXT UNIQUE NOT NULL);
CREATE TABLE
```

exercise-# \d students

Table "public.students"				
Column	Type	Collation	Nullable	Default
student_id	integer		not null	nextval('students_student_id_seq'::regclass)
name	text		not null	
email	text		not null	

Indexes:

"students_pkey" PRIMARY KEY, btree (student_id)

"students_email_key" UNIQUE CONSTRAINT, btree (email)

Q2. Create a table having all the DataTypes in PostgreSQL

```
Query  Query History
1  CREATE TABLE datatype_table(
2      id SERIAL PRIMARY KEY,
3      name TEXT ,
4      age INTEGER,
5      salary NUMERIC(10,2),
6      is_active BOOLEAN,
7      created_at TIMESTAMP,
8      profile JSON,
9      bit_col BIT(8)
10 )
```

id	name	age	salary	is_active	created_at	profile	bit_col
[PK] integer	text	integer	numeric (10,2)	boolean	timestamp without time zone	json	bit

Q3. Create a table with primary key constraints

```
Query  Query History
1  CREATE TABLE students (
2      student_id SERIAL PRIMARY KEY,
3      first_name TEXT NOT NULL,
4      last_name TEXT NOT NULL,
5      email TEXT UNIQUE NOT NULL,
6      admission_date DATE NOT NULL
7  );
```

	student_id [PK] bigint 	first_name text 	last_name text 	email text 	admission_date date 
--	---	--	---	---	--

Q4. Run all basic CRUD queries on the table

CRUD OPERATIONS:

Create:

Query	Query History
1	CREATE TABLE students (2 student_id SERIAL PRIMARY KEY , 3 first_name TEXT NOT NULL , 4 last_name TEXT NOT NULL , 5 email TEXT UNIQUE NOT NULL , 6 admission_date DATE NOT NULL 7);

Create (Insert):

Query	Query History
1	INSERT INTO students (first_name, last_name, email, admission_date) 2 VALUES 3 ('Harsh', 'Raj', 'harsh@email.com', '2026-02-01'), 4 ('Aman', 'Mittal', 'aman@email.com', '2026-02-15');

Read (Select):

```
Query  Query History
1      SELECT * FROM students
```

	student_id [PK] integer	first_name text	last_name text	email text	admission_date date
1	1	Harsh	Raj	harsh@email.co...	2026-02-01
2	2	Aman	Mittal	aman@email.co...	2026-02-15

Update:

```
1      UPDATE students
2      SET last_name = 'Pandey',
3          admission_date = '2025-03-02'
4      WHERE student_id = 2;
```

Delete:

```
1      DELETE FROM students
2      WHERE student_id = 2;
```

Q5. Implement WHERE, GROUP BY, ORDER BY, HAVING in filters.

First creating and inserting values into a table to perform these tasks.

```
1 CREATE TABLE sales (  
2     sale_id SERIAL PRIMARY KEY,  
3     product_name TEXT NOT NULL,  
4     category TEXT NOT NULL,  
5     quantity INTEGER NOT NULL,  
6     price NUMERIC(10,2) NOT NULL,  
7     sale_date DATE NOT NULL  
8 );
```

```
1 INSERT INTO sales (product_name, category, quantity, price, sale_date)  
2 VALUES  
3 ('Laptop', 'Electronics', 2, 50000, '2024-01-10'),  
4 ('Mobile', 'Electronics', 5, 20000, '2024-01-12'),  
5 ('Shirt', 'Clothing', 10, 1500, '2024-01-15'),  
6 ('Jeans', 'Clothing', 4, 2500, '2024-01-18'),  
7 ('TV', 'Electronics', 1, 60000, '2024-01-20');
```

WHERE CLAUSE:

```
1 SELECT * FROM sales  
2 WHERE category = 'Electronics';
```

	sale_id [PK] integer	product_name text	category text	quantity integer	price numeric (10,2)	sale_date date
1	1	Laptop	Electroni...	2	50000.00	2024-01-10
2	2	Mobile	Electroni...	5	20000.00	2024-01-12
3	5	TV	Electroni...	1	60000.00	2024-01-20

GROUP BY CLAUSE:

```
Query  Query History
1  SELECT category, SUM(quantity) AS total_quantity
2  FROM sales
3  GROUP BY category;
```

	category 	total_quantity 
1	Electroni...	8
2	Clothing	14

HAVING CLAUSE:

```
Query  Query History
1  SELECT category, SUM(quantity) AS total_quantity
2  FROM sales
3  GROUP BY category
4  HAVING SUM(quantity) > 9;
```

	category 	total_quantity 
1	Clothing	14

